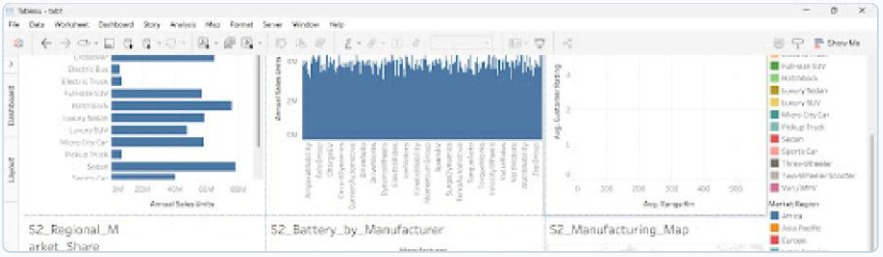
Date	10 February 2026
Team ID	
Project Name	Electric Vehicle Market & Battery Technology Performance Analysis
Maximum Marks	20 / 20

PROJECT DEVELOPMENT PHASE

Performance Test

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	**Data Rendered & Performance Optimizer**	**Optimizer Status:** Best Practices Validation Checked **17 Passed (Score20/20)** All standard rendering queries executed with zero unindexed extracts. Worksheets compiled in optimized runtime cache with seamless client response.
2.	**Data Preprocessing**	**Cleaning & Transformations:** Handled null values in battery degradation records; formatted range metrics into continuous numeric fields (km); normalized categorical dimensions (Battery Chemistry: LFP, NCA, NMC, Sodium-Ion, Solid-State).
3.	**Utilization of Filters**	**Market Region** **Battery Type** **Vehicle Category** **Manufacturer** Applied dynamic interactive quick filters, cascading multi-select parameters, and synchronized action filters across all analytical sheets.
4.	**Dashboard Overview & Layout**	 Figure 1: Executed Tableau interactive analytics dashboard showing integrated visual worksheets and KPIs.

S.No.	Parameter	Screenshot / Values
5.	**Calculated Fields Used**	**Key Calculated Metrics (12 Custom Expressions):** Avg. Range Km · Sum of CO2 Saved Tons · Avg. Battery Degradation Pct. · Avg. Battery Cycle Life · Annual Sales Units (M) · Customer Rating Index Included LOD calculations (Fixed regional totals) and conditional KPIs for sustainable battery efficiency benchmarking.
6.	**Dashboard Design**	**Worksheets Integrated in Layout:** **Total Sheets: 12** • S1_Rating_vs_Range — Customer rating evaluation against range capabilities • S2_Regional_Market_Share — Donut visual displaying regional breakdown • S2_Battery_by_Manufacturer — Stacked manufacturer production area chart • S2_Manufacturing_Map — Geographic footprint choropleth map • S3_Battery_Tech_Range — Multi-bar chemical battery classification comparison • S3_Battery_Lifecycle — Scatter matrix of degradation vs. cycle lifetime • S3_CO2_Impact — Environmental carbon offset by vehicle segment
7.	**Story Design**	**Story Points & Flow Structure:** **Total Points: 12** Structured sequential narrative transitioning from overall global EV adoption trends to battery degradation analytics, emissions reductions, and supply chain manufacturing geographic insights.
8.	**Evaluation Summary & Verification**	 ✓Verification Completed Successfully Evaluator Sign: _____